AA-D-2757
May 16, 1990

FEDERAL SPECIFICATION

DOOR, VAULT, SECURITY

This Federal Specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1 SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers security, vault doors as required in the Information Security Oversight Office, Directive No. 1, National Security Information. The vault doors provide protection against unauthorized entry for the periods of time specified in 1.2.1.

1.2 Classification.

1.2.1 Classes. The vault doors under this specification shall be of the following class.

Class 8 - Resistant to 15 minutes forced entry, 30 man minutes covert entry and 20 hours surreptitious entry.

1.2.2 Types. The vault doors shall be of the following types as specified (see 6.2).

Type I - Single leaf door.

Type II - Single leaf door, no exterior hardware.

Type III - Double leaf door.

1.2.3 Design. The vault doors shall be of the following designs, as specified (see 6.2). For double leaf doors, the design shall apply to the leaf in which the lock is mounted.

Design R - Right opening swing (right hand reverse).

Design L - Left opening swing (left hand reverse).

1.2.4 Styles. The vault doors shall be of the following styles as specified (see 6.2).

Style 1 - With an optical device.

Style 2 - Without an optical device.

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2 APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bids or request for proposals, shall apply.

Federal Specification:

FF-L-2740 - Locks, Combination.
QQ-C-320 - Chromium plating (Electrodeposited).
QQ-P-416 - Plating, Cadmium (Electrodeposited).
TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.
PPP-C-650 - Crates, Wood, Open and Covered.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment

Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Specifications and Standards and at the price indicated in the Index. The index which included cumulative monthly supplements, as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

(Single copies of this Specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta,, Chicago, Kansas City, Fort Worth, Denver, San Francisco, Los Angeles and Seattle.

(Federal Government activities may obtain Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Sample panels of the standard colors are obtainable, without charge, from the Business Service Center, Federal Supply Service, General Services Administration, Washington, DC 20407, or from the Business Service Center of the nearest Regional

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Office.)

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal, shall apply.

American Society for Testing and Materials (ASTM):

B-633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1616 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, NW, Washington DC, 20036.)

Underwriters Laboratories, Inc (UL) Publications:

UL 608 - Burglary Resistant Vault Doors and Modular Panels.
UL 437 - Key Locks

(Application for copies should be addressed to Underwriters Laboratories, 333 Pfingsten Rd., Northbrook, IL 60062.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3 REQUIREMENTS

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3.1 Qualification. The vault doors furnished under this specification shall be products which have been tested and have passed the qualification tests and inspections specified in section 4, and have been listed on or approved for listing on the applicable qualified products list (QPL).

3.1.1 Qualification suspension.

3.1.1.1 Development of entry techniques. The doors qualified under this specification will be continually tested by the Government during the term of qualification to determine whether the surreptitious or forced entry protection afforded by the doors should or can be improved. If at any time, entry techniques are developed within the framework of the specification which affect a door's security integrity, it shall be removed from the QPL.

3.1.1.2 Change in specification requirements. This specification will be reviewed by the Government to determine whether the specification requirements should or can be changed to improve product quality. If, at any time, requirements are changed and such changes affect the qualification status of a qualified door, it shall be removed from the QPL and the manufacturer will be required to modify the product to the extent necessary to comply with specification changes and have the product requalified.

3.2 Materials. Material used in the door's construction shall be as specified herein. Material not specified shall be of good commercial quality, suitable in all respects for the purpose intended.

3.2.1 Steel. Steel used in the door shall be type, thickness and strength to meet all applicable requirements of this specification. Steel shall be free from rust, scale, pits, buckles and other imperfections which might adversely affect the appearance or the serviceability of the finished product.

3.2.2 Face hardware. The face hardware, excluding combination locks, shall be satin finished anodized aluminum, type 430 corrosion resistant steel or satin finished chromium plating on steel or on die-cast zinc, brass or bronze. The exposed surfaces of all hardware used on a single unit shall be finished to match each other within the limits of the base material and protective coating used. The exposed surfaces of all face hardware shall be free of sharp edges, burrs, pits, nicks or scratches that penetrate the protective plating or anodizing.

3.2.3 Finishing materials.

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3.2.3.1 Enamel and lacquer. The final coat for the door shall be either baked enamel, air dried, textured finish, nitrocellulose lacquer or water reducible coating. The quality of the final coat and its application shall be in accordance with good commercial standards and practices. The color shall be as specified in 3.2.4.

3.2.3.2 Chromium plating. Chromium plating shall be in accordance with class I, type II of QQ-C-320.

3.2.3.3 Cadmium plating. Cadmium plating shall be in accordance with class I of QQ-P-416.

3.2.3.4 Zinc plating. Zinc plating shall be in accordance with ASTM B-633.

3.2.4 Color of finish. The color of finish shall be gray, color no. 26134, of Federal Standard No. 595.

3.3 Construction and design.

3.3.1 UL compliance. Vault doors shall meet the class M requirements of UL 608. Proof of compliance with the UL standard shall be provided with the qualification request. Proof shall consist of test reports from a nationally recognized testing laboratory which shows that the testing has been conducted and passed on the door design offered to the Government. The UL label will be accepted as proof of compliance.

3.3.2 Design. The vault door design shall provide an end product that is acceptable in general appearance. The door shall be hinged to swing, as specified.

3.3.3 Assembly. The door assembly shall consist of the vault door and the door frame. The frame shall be considered a part of the door for the purpose of entry resistance testing and shall afford the same security protection as the door. Except that hinges shall be removable from the outside, the door shall be assembled in such a manner as to preclude the removal or loosening of any of the door's components when the door is closed and locked. The door and frame shall be in perfect alignment and operation of the locking mechanism, including the locking bolts, shall be smooth and positive without binding or jamming of parts. The door and frame shall interlock or be of such design that there is no clear path between the exterior and the interior of the door. The door shall withstand the test in 4.4.8.1.

3.3.4 Dimensions.

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3.3.4.1 Single leaf doors. The vault door assembly shall be of one size, and when installed, shall have a clear door opening of 78 inches high and 40 inches wide. A manufacturing tolerance of ± 1/8 inch shall be permitted.

3.3.4.2 Double leaf doors. The vault door assembly shall be of one size, and when installed, shall have a clear door opening of 78 inches high and 80 inches wide. Doors shall be of equal width.

3.3.5 Door frame. The single leaf door frame shall be 60 inches wide and 88 inches high. The double leaf door frame shall be 100 inches wide and 88 inches high. The door frame shall be non-grout type. Protection for the extended locking bolts shall be built into the door frame. The frame shall be designed so that when attached to the wall, the wall clamping bolts will be exposed only to the inside of the vault. The frame shall have leveling and adjusting screws to compensate for building sag which may occur at any time in the future.

3.3.6 Wall opening. The single leaf door assembly shall be designed to mount in a wall opening 48 inches wide and 82 inches high. The double leaf door assembly shall be designed to mount in a wall opening 88 inches wide and 82 inches high. The assembly shall be capable of adapting to wall openings which vary ± 1 inch from the specified opening. The door assembly shall mount in one of the following nominal wall thickness, as specified (see 6.2): 6 inch; 8 inch; 10 inch; or 12 inch. The assembly shall provide a ± 1/2 inch adjustment to allow for variation in the nominal wall thickness.

3.3.7 Door pull and throw bolt handles. The door pull and throw bolt handles shall be of a material specified in 3.2.2. They shall be not less than 4 inches in length and of designs consistent with their intended uses. The handles shall be without burrs, nicks, scratches and sharp edges. They shall be securely and firmly attached to the door to withstand loosening in testing and in operation during the service life of the door. The door pull handle may be integral with the throw bolt handle. Removal of the handle arbor shall be controlled only from inside of the door. The throw bolt handle shall require not more than 15 foot-pounds of torque to engage or disengage the bolt mechanism.

3.3.8 Door stop. A door stop to prevent the door's face hardware from striking wall surfaces shall be furnished with the door. The stop shall be designed to be mounted on a wall or floor and not on the door. The stop shall be able to withstand hard usage. The stop shall not scratch or scar the door's painted finish, when the door is swung open against the stop.

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3.3.9 Door striker. The door shall have a striker on both the front and hinged edges to minimize play or shake in the door when in the locked condition. The fit of the door to the striker on both edges shall be such that there is not more than 1/32 inch play or shake in the door when the bolts are thrown to the locked position.

3.3.10 Door hinges. The door shall be mounted to the frame by not less than three anti-friction, bearing hinges, designed to allow the door to be opened approximately 180 degrees. The hinges shall be removable. The bolts for the door side of the hinge shall be accessible from the inside of the door. The bolts for the frame side of the hinge shall be accessible from the exterior. The maximum force required to swing the unlocked door from any position shall not exceed 10 pounds, when tested in accordance with 4.4.8.2.

3.3.11 Door threshold. The door threshold shall be designed to provide a ramp of approximately 1/4 inch at the door threshold to permit free swing of the door after its erection. If receptive cups, ports or grooves are used, they shall be recessed not less than 1/2 inch below the bolt, in its extended position, to prevent dirt or other substances from obstructing the locking mechanism.

3.3.12 Back cover plate. A back cover of not less than 0.0598 inch steel shall completely enclose the back of the door and shall be in three parts. The back plates shall be firmly and securely fastened to the door and shall be reinforced or attached by a method to prevent bowing or distortion. The back cover shall be easily removed for service and inspection purposes. The removal and replacement of the back cover shall be accomplished by one person using common hand tools. The back plate shall have an opening covered by a sliding service door. The opening, with the service door opened, shall be large enough and positioned so as to allow maintenance of the door's combination lock. The service door may also be used to access the escape device. When the service door is open, it shall not be possible to secure the vault door. The service door shall have a key lock meeting U.L. Standard UL 437 for a security container key lock, to prevent tampering. A minimum of two keys shall be provided.

3.4 Changeable combination lock. A changeable combination lock in accordance with Federal specification FF-L-2740 shall be installed as an integral part of the vault door. The lock shall be a Class TR, Type NT, Size LD lock, and shall be hand change, Model HC, or key change, Model KC, as specified (see 6.2).

3.4.1 Lock installation. The lock shall be mounted in accordance with the manufacturer's recommended practice. The

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lock, excluding the lock bolt, shall not be altered in any manner from the formation supplied by the lock manufacturer. The lock's dial ring shall be flat against the front of the door face, and its attachment to the door shall be firm and secure without movement or side play. The lock case shall be protected by a lock box. The lock case shall be securely attached with screws and locks washers. There shall be no movement or side play to the lock case. The plates, housing or other barriers used around the lock case shall be installed in a manner so as not to abrade or otherwise damage the lock.

3.4.2 Lock box. The lock box shall be mounted to the door in such a manner as to provide a space between the lock box and the door. The space shall allow insertion of additional plates of steel or other materials, up to 1/2 inch thick.

3.5 Locking mechanism. The door lock bolts shall be of a design, size and material strength to withstand the applicable tests in section 4. A minimum of six live bolts shall be used. The bolts shall operate easily and smoothly, without binding or jamming. The bolts shall not dent or otherwise deface the door frame in their movement. The attaching linkages shall be channeled, strapped or welded. The locking mechanism shall have a detent to hold the bolts in the open position, when the bolts are retracted and the door swung open. The mechanism shall allow the combination lock to be secured with the bolts retracted. The detent release shall be designed to automatically throw the bolts when the door is closed, but prevent inadvertent tripping. On double leaf doors, the detent shall not extend the bolts, unless the passive leaf is closed. If the lock is secured while the door is open, when the door is closed, the bolts shall be automatically thrown, and the door shall withstand the entry resistance tests.

3.6 Escape device. Each vault door shall have an escape device which shall be permanently installed. The device shall permit ready egress for persons locked inside the vault area. The escape device shall be designed such that after it is exposed, it requires two, unidirectional actions, such as push and turn, to activate the release of the vault door bolts. Access to the device shall only be from the inside of the vault, and its design shall be such that under normal operating conditions it cannot be activated from the outside. A decal shall be permanently affixed to the inside face of the door frame outlining in easily read letters, complete, understandable instructions for activating the device to open the door. Neither the design of the device nor its installation or use shall affect the door's resistance to entry techniques.

3.7 Optical device. Style 1 vault doors shall have a wide

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angle optical device located five feet above the floor and centered in the door. If the device cannot be centered, due the door design, it shall be located as close to the center as possible. In no case shall the device be located closer than eight inches from the edge of the door. The optical device shall not be removable from the exterior of the door without damage.

3.8 Lubrication. The door's moving parts requiring lubrication shall have a permanent type lubricant applied which is suitable to varied climatic conditions likely to be encountered during service of the product.

3.9 Pretreatment and finish.

3.9.1 Pretreatment. All exterior and interior ferrous metal surfaces shall be treated for painting in accordance with any type in TT-C-490. Special attention shall be given to the door's interior to assure that all welds are clean and that all slag. splatter and dirt accumulation is removed.

3.9.2 Finish. The final coat used for the finish shall be as specified in 3.2.3.1 and it shall be applied to all exterior and interior metal surfaces except plated metal. The minimum total finished film thickness of the final coat shall be not less than 1.0 mil. The finish shall level out to produce uniform exposed surface without runs, wrinkles, grit, areas of thin or no film or separation of color. A textured or crinkle finish may be used. Special attention shall be given to ensure that all surfaces are protected against rust. The final shall adequately withstand the test in 3.9.2 without evidence of cracking, flaking or loss of adhesion of the finish.

3.9.3 Plating. Bolts, screws, nuts and similar hardware shall be made to resist rust by electrogalvanizing or by zinc, chromium or cadmium plating as specified in 3.2.3.

3.10 Resistance to entry.

3.10.1 Forced, covert and surreptitious entry. The vault door shall resist entry for the periods of time specified in 1.2.1 when tested in accordance with 4.4.8.3.

3.10.2 Radiology. The vault door and its locking mechanism shall resist entry for not less than 20 hours, when tested in accordance with 4.4.8.4.

3.11 Labels. Each door furnished under this specification shall bear the metal labels specified hereunder.

3.11.1 General Services Administration label. The label

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shall be affixed to the outside face of the door and shall show in letters not less than 1/8 inch high, the following.

GENERAL SERVICES ADMINISTRATION
APPROVED SECURITY VAULT DOOR
MANUFACTURER'S NAME

3.11.2 Identification label. Affixed to the inside face of the door frame shall be a label which shall show the door model and serial number, date of manufacture, and Government contract number.

3.11.3 Certification label. Affixed to the inside face of the door frame shall be a label which shall bear the following certification.

> This is a U. S. Government, Class 8 vault door which has been tested and approved by the Government under Fed. Spec. AA-D-0000. It affords the following security protection:
>
> 15 minutes against forced entry.
> 30 minutes against covert entry.
> 20 hours against surreptitious entry.
>
> The protection certified above applies only to the door and not to the vault proper.

3.11.4 External identification label. All vault doors furnished under this specification shall have a number label securely affixed to the front face of the product by two zinc plated drive screws. The label attachment shall not degrade the door's security protection. The label shall be mounted on the door frame, centered between the left and right edges and between the top of the door and the top of the frame. The label shall be 0.020 inch thick, satin finished aluminum and shall be 2-1/2 inches wide by 11/16 inch high. The label numbering system shall be established by the manufacturer to provide non-repetitive numbers. The label numbers shall be 1/4 inch high and shall be embossed or engraved.

3.12 Workmanship. The workmanship shall be of a quality to produce a finished end item which is able to withstand hard daily usage. The edges of all exposed parts and sheets shall be protected by folding, beading, flanging or grinding to eliminate burrs, roughness and sharp edges. The bending of channels and flanges shall be straight and smooth. Welding and brazing shall produce rigid and secure connections. All welding and brazing shall be sound without porosity and joints shall be in proper alignment. All protruding or depressed welds on the doors

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exterior surface shall be filled and sanded or ground smooth. Lock washers, cotter pins, clips and other retainers or built-in features shall be used to prevent loosening of screws, bolts and nuts which may cause disengagement of parts and possible lockout. Moving parts shall operate smoothly without binding or jamming. To assure strict compliance with 3.4.1, particular attention shall be given to the quality of workmanship and method used in the installation of the combination lock in the door. The door shall be free of any defects or features which may adversely affect its appearance and serviceability or which may cause injury to personnel.

3.13 Replacement of parts. Parts subject to replacement, such as the combination lock and face hardware, shall be capable of identical replacement in the field without the use of special tools or specially qualified personnel. Such replacements shall be possible without affect the integrity of the door.

3.14 Spare parts list. A spare parts list or all door components which may be subject to subsequent replacement shall be furnished with each door. The parts list shall clearly identify the parts by description and part numbers. The list shall be printed on paper or other suitable material and bonded by adhesive to the inside face of the door frame.

3.14.1 Locking mechanism and lock mounting drawings. Complete, exploded view drawings of the locking mechanism and the lock mounting, with individual parts identified, shall be furnished by the manufacturer upon specific request of the purchaser.

4 QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. Except that testing for qualification shall be performed by an agency designated by the General Services Administration, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may use his own or any other inspection facility or service, acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government, as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in

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accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with the specification.

4.3 Inspection of preparation for delivery. An inspection shall be made to determine that the packing and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with Table I. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of items in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects.

Examine	Defect
Markings (exterior)	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Incomplete closure of box, loose strapping, distortion of container.

4.4 Testing procedures and tests.

4.4.1 Qualification testing agency. Qualification tests accomplished on doors submitted for approval for inclusion on the applicable Qualified Products List (QPL) and any retesting that may be required shall be performed by a testing agency specifically designated by the General Services Administration.

4.4.2 Test costs. All testing costs entailed in determining the qualification of the supplier's product, including the cost of retesting a qualified product is subsequently disqualified under 3.1.1, shall be borne by the supplier, and shall be payable to the General Services Administration as directed by the Furniture Commodity Center, Federal Supply Service.

4.4.3 Test procedures. The following procedures shall govern the testing of all doors submitted for qualification under this specification.

(a) Samples shall be submitted for qualification only after the supplier has obtained written authorization from

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the General Services Administration.

(b) A qualification test may be discontinued by the Government at any time the product fails to meet one or more of the requirements set forth in this specification. The manufacturer may be permitted to make modification to the sample during the testing phase where such modification, in the judgement of the General Services Administration and the testing facility, are clearly in the best interest of the Government.

(c) In case of failure of a sample, consideration will be given to the request of the manufacturer for resubmission for retest only after it has been clearly shown that changes have been made in the product which the Government considers sufficient to warrant retest.

(d) The manufacturer or his representative will not be permitted to observe the actual testing conducted on his product at the testing facility. However, when samples tested fail to comply with the requirements of this specification, the sample may be examined by the manufacturer or his representative and full details of the failure may be made known to them in a manner which for reasons of security, will be in the best interest of the Government.

4.4.4 Test samples. A qualification test sample of the class the supplier proposes to furnish shall be forwarded at a time and to a place designated by GSA. In the event the sample is destroyed or damaged to such an extent during testing that testing cannot be completed, the Government reserves the right to require the manufacturer to furnish additional samples to complete the testing. Two panels of 0.0359 inch steel shall be furnished with the sample door for the purpose of the bend test. One panel shall be prepared to reflect the inner coating and one to reflect the outer coating.

4.4.5 Drawings and list of material. The manufacturer shall furnish five complete set of construction and assembly drawings and lists of materials with samples submitted for qualification. When the samples are tested and are approved for inclusion on the applicable QPL, three sets of drawings and list of materials will be approved by the Government for use in inspection of products offered under contract. All material so furnished by the manufacturer will be held in proprietary confidence.

4.4.5.1 Changes in construction or drawings. Once a product has been tested and approved for QPL, no subsequent change of any

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kind shall be made in its construction or in the construction drawings unless prior written authorization to make a change is obtained by the manufacturer from the Federal Supply Service, GSA.

4.4.6 Qualification testing and inspection. Qualification testing and inspection shall consist of the following tests and inspections. Failure to meet any one or more of these requirements shall provide reason to consider the product as having failed to meet the requirements for qualification.

(a) Rack test - 4.4.8.1
(b) Door operating force - 4.4.8.2
(c) Forced, covert and surreptitious entry resistance - 4.4.8.3
(d) Radiology test - 4.4.8.4
(e) Bend test - 4.4.8.5
(f) Inspection - 4.4.9

4.4.7 Inspection and testing for acceptance. The Government reserves the right to inspect and test each door, including all component parts thereof, delivered for acceptance under this specification after award of a contract.

4.4.8 Test methods.

4.4.8.1 Rack test. The vault door shall be mounted in a test frame. The test frame opening shall be set to the maximum dimensions established by the manufacturer, for the use of the door. The door shall be swung open 90 degrees from its closed position. Two hundred pounds shall be loaded on the top edge of the door opposite and furthermost from the hinge side. The door shall be allowed to hang in this position for a period of 24 hours. At the end of this period, the door shall be examined for ease of operation. The door shall not bind or jam in its frame and the lock and locking mechanism shall operate easily and smoothly.

4.4.8.2 Door operating force. After the rack test, the vault door shall be closed completely, with the door handle restrained to prevent extension of the locking bolts. The force required to open the door to approximately 90 degrees shall be measured. Force shall be applied gradually until the door begins motion. Force required to maintain the door in motion at a rate of 30 to 45 degrees/second shall be measured. The door shall be stopped at 90 $\pm$ 10 degrees, and the force required to begin and maintain the motion to complete closure shall be measure.

4.4.8.3 Forced, covert and surreptitious entry resistance. There shall be sufficient time and opportunity to study the

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design and construction of the vault door and to develop testing methods prior to the start of testing. There shall be no limit to the number of methods of surreptitious, covert and forced entry attempted. Not more than two men shall be used simultaneously during each attempt at entry. The man-minute working time shall cover the period during which a surreptitious or forced entry test on the door is in progress and shall be exclusive of time required for safety precautions and rest periods.

4.4.8.3.1 Tools and devices. Tools and devices used in the entry tests are limited as specified in UL 608, except that the weight of the tools used for a single test shall not exceed 150 pounds. Tools and devices shall be capable of being carried in two cases or bags, each case or bag not exceeding 1.5 cubic feet in volume. The total weight of the tools used in a single test shall not exceed 150 pounds, exclusive of the weight of the case. The test tools and devices selected for a particular attempt shall be weighed prior to commencement of the test.

4.4.8.3.2 Timing. The time clock shall be started when the test equipment is picked up to approach the sample and shall not be stopped during the test except as specified above. Any change or repair of tools taken from the carrying case during a test shall only be done while the clock is running. The tests must be conducted in a manner that is repeatable. Any surreptitious, covert or forced entry through the vault door under the above conditions, within the time specified for the door's class, shall provide reason to consider the door as having failed to meet the requirement.

4.4.8.4 Entry by radiological techniques. The vault door shall successfully meet the following test to demonstrate resistance to entry by radiological techniques. The vault door structure shall be radiographed and the resulting radiographs shall not permit determination of the lock combination to the extent that entry is made into the vault in less than the time specified. Radioactive isotopes and other sources, of any type judged to be effective for the purpose of this test, will be used. Any effective radiation shielding provided in the vault door will be included in the test. The test is intended to simulate attempted entry within the specification limit of 150 pounds of equipment, utilizing practical and feasible procedures and equipment available to Government testing agencies performing the tests. Any entry made under the preceding conditions within 20 man-hours shall be considered a failure of the vault door to meet the requirement of this specification.

4.4.8.5 Bend test. The sample panels shall at room temperature be bent around a 1/4 inch rod to an angle of 180 degrees and then examined for compliance with 3.9.2.

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4.4.9 Inspection. A visual inspection shall be made of the vault door to determine compliance with the requirements specified in Section 3 of the specification.

5 PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.1.1 Level A. Each complete door shall be packed in a crate conforming to PPP-C-650 or in a box conforming to PPP-B-621. When packed in a crate, the contents shall be waterproof shielded with a shroud fabricated of material conforming to PPP-B-1055, as specified for interior shrouds, and in accordance with the appendix to the crate specification. When packed in a box, the contents shall be shrouded as specified herein and the box shall be modified with the addition of reinforcing members and skids as specified in the box specification. The contents of the box or crate shall be blocked, braced and cushioned to prevent movement during multiple shipments.

5.1.2 Level B. Each complete door shall be packed in a crate conforming to PPP-C-650 or in a box conforming to PPP-B-621. The contents of the box or crate shall be blocked, braced and cushioned to prevent movement during multiple shipments.

5.1.3 Level C. Each door shall be packed to assure acceptance and safe delivery to destination in containers in compliance with the National Motor Freight Classification and the Uniform Freight Classification.

5.2 Marking. Marking shall be in accordance with 5.2.1 or 5.2.2, as specified (see 6.2) and shall include all precautionary marking required by the National Motor Freight Classification and the Uniform Freight Classification for items as described in the specification.

5.2.1 Civil agencies. Each shipping container shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. Each shipping container shall be marked in accordance with MIL-STD-129.

6 NOTES

6.1 Intended use. The doors are intended for use in storage vaults and strongrooms to protect against unauthorized passage of a person or persons through the doorway into the vault.

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6.2 Ordering data. Purchasers should exercise any desired option offered herein, and procurement document should specify the following.

(a) Title, symbol and date of this specification.
(b) Class, type, design and style required (see 1.2).
(c) Thickness and composition of vault wall (see 3.3.6 and 6.4).
(d) Level of packing required (see 5.1).
(e) Marking required (see 5.2).

6.3 Qualification. With respect to the products requiring qualification, awards will be made only for such products as have at the time set for opening of bids, been tested and approved, for inclusion on the QPL, whether or not such products have actually been listed by that date. The attention of suppliers is called to this requirement, and the manufacturers are urged to arrange to have products that they propose to offer the Federal government tested for qualification so that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the General Services Administration, Federal Supply Service, Furniture Commodity Center, Engineering Division, Washington, DC 20406, and information pertaining to qualification may be obtained from that activity.

6.4 Composition of vault wall. Examples of materials commonly used in vault construction are reinforced concrete, interlocked hard brick, steel alloy or a combination of these. In order that the door manufacturer can insure a proper fit, the purchaser should stipulate in the purchase order, the thickness and type of material used in the vault wall.

6.5 Definitions of terms used in this specification.

6.5.1 Entry. For the purpose of this specification, entry means: (1) opening the door, (2) creating an opening to allow passage through the vault door or wall opening in which the door is mounted.

6.5.2 Forced entry. For the purpose of this specification, forced entry means a method of entry which would leave evidence of the act which could be readily discernible in the normal use of the door. Forced entry is considered to be an attack in which the attacker has no concern over leaving evidence that the vault door has been penetrated.

6.5.3 Covert entry. For the purpose of this specification, covert entry means a method of entry which would leave evidence of the act which would not be detectable in the normal use of the

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door but which could be detected by a qualified person during inspection.

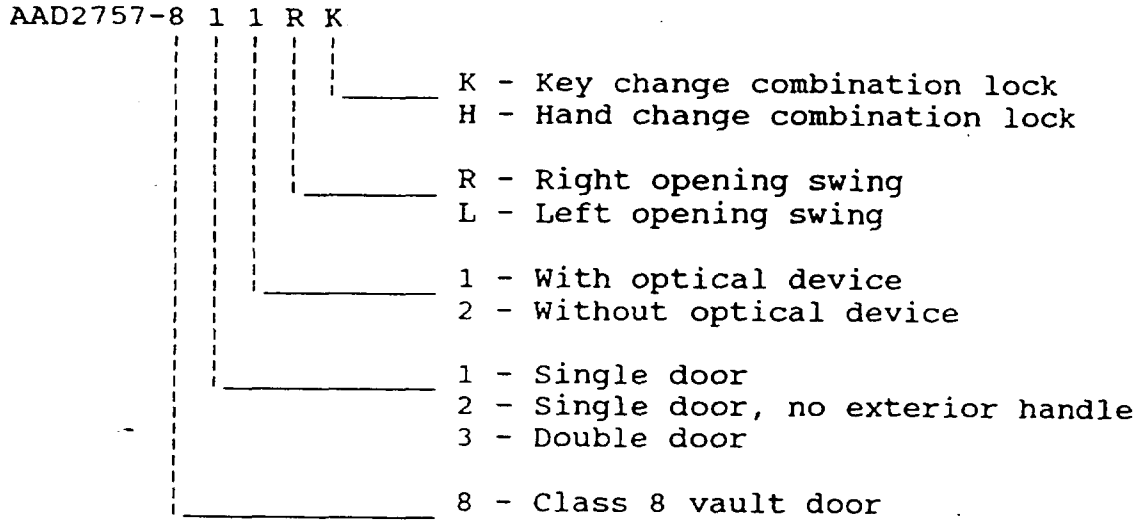
6.5.4 Surreptitious entry. For the purpose of this specification, surreptitious entry means a method of entry, such as lock manipulation or radiological attack on the combination lock, which would not be detectable during normal use or during inspection by a qualified person.

6.5.5 Normal use. For the purpose of this specification, normal use means the opening of the combination lock, releasing the locking mechanism, opening the door to the extent necessary for entry and closing and relocking the door. During normal use, it is considered the sides, front, back and threshold of the door are exposed to view and touch. The top and bottom edged of the door are not considered to be exposed to view or touch.

6.6 Samples. All samples required for test purposes shall be furnished at no expense to the Government, and the manufacturer shall pay all transportation to and from the point where the tests are performed. All tested samples shall become property of the Government but may be released to the manufacturer at the option of the Government. Upon request, the manufacturer shall furnish to the Government testing facility, a door equal in all respects to that of the qualified sample for use in inspection and test during the terms of qualification. The door shall be furnished at no expense to the Government and will be returned to the manufacturer upon removal of his product from the qualified products list.

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6.7 Reference identification number. The reference identification number (RIN) system may be used for items covered by this specification. An example of the RIN is as follows.



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